01/31/2022

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M168_Notes_2.5

2.5 Multiplication of Matrices

Example

A flower shop sells 96 roses, 250 carnations and 130 daisies in a week. The roses sell for \$2 each, the carnations for \$1 each and the daisies for \$0.50 each. Find the revenue of the shop during the week.

$96\times 2 + 250\times 1 + 130\times 0.50 = 507$

Express the number of flowers in a 1×3 matrix:

$$\begin{bmatrix} 96 & 250 & 130 \end{bmatrix}$$

Next express the price as a 3×1 matrix:

$$\begin{bmatrix} 2 \\ 1 \\ 0.5 \end{bmatrix}$$

$$\begin{bmatrix} 96\times 2 + 250\times 1 + 130\times 0.5 \end{bmatrix} = \begin{bmatrix} 507 \end{bmatrix}$$

In general, if A is $1\times n$ and B is $n\times 1$, the product AB is a 1×1 matrix:

$$A\times B = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \end{bmatrix} \times \begin{bmatrix} b_{11} \\ b_{21} \\ \vdots \\ b_{n1} \end{bmatrix} = \begin{bmatrix} a_{11}\times b_{11} + a_{12}\times b_{21} + \cdots + a_{1n}\times b_{n1} \end{bmatrix}$$

$$B = \begin{bmatrix} b_{11} \\ b_{21} \\ \vdots \\ b_{n1} \end{bmatrix}$$

$m\times n$ $\quad r\times p$

if $n = r$

If A is an $m\times n$ matrix and B is a $n\times p$ matrix, then the product matrix $A\times B = C$ is an $m\times p$ matrix.

$$A\times B = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & & a_{mn} \end{bmatrix} \begin{bmatrix} b_{11} & b_{12} & \cdots & b_{1p} \\ b_{21} & b_{22} & \cdots & b_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ b_{n1} & b_{n2} & \cdots & b_{np} \end{bmatrix} = C \qquad m\times p$$

$$= \begin{bmatrix} a_{11}\times b_{11} + a_{12}\times b_{21} + \cdots + a_{1n}\times b_{n1} & (ab)_{12} & \cdots & (ab)_{1p} \\ (ab)_{21} & (ab)_{22} & \cdots & (ab)_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ (ab)_{m1} & (ab)_{m2} & \cdots & (ab)_{mp} \end{bmatrix}$$

$$c_{11} = \sum_{j=1}^{n} a_{1j}\, b_{j1} = a_{11}\cdot b_{11} + a_{12}\cdot b_{21} + \cdots + a_{1n}\cdot b_{n1}$$

$$c_{ij} = \sum_{k=1}^{n} a_{ik}\, b_{kj}$$

Matrix multiplication is not commutative. In general, $AB \neq BA$

Example
Find the products AB and BA where

$$A = \begin{bmatrix} 1 & 0 \\ 2 & 3 \end{bmatrix} \text{ and } B = \begin{bmatrix} 1 & 2 \\ 0 & -3 \end{bmatrix}$$

$$A \cdot B = \begin{bmatrix} 1 & 0 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 0 & -3 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 2 & 4-9 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 2 & -5 \end{bmatrix}$$

$$B \cdot A = \begin{bmatrix} 1 & 2 \\ 0 & -3 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 2 & 3 \end{bmatrix} = \begin{bmatrix} 1 \cdot 1 + 2 \cdot 2 & 1 \cdot 0 + 2 \cdot 3 \\ 0 \cdot 1 + -3 \cdot 2 & 0 \cdot 0 + (-9) \end{bmatrix} = \begin{bmatrix} 5 & 6 \\ -6 & -9 \end{bmatrix} \neq \begin{bmatrix} 1 & 2 \\ 2 & -5 \end{bmatrix} = AB$$

(2,1) (2,2) (1,2)

Means $AB \neq BA$.

(4×3) (2×4) (3×4) $(m \times n) \times (n \times p) \rightsquigarrow (m, p)$

One special matrix is called the identity matrix, I.
It is a square matrix with 1's on the diagonal and zeros elsewhere,

$$I = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix}$$

I_2 is a 2×2 identity matrix and I_n is an $n \times n$ identity matrix.

```
NAMES EDIT      identity(3)
1:det(                [[1 0 0]
2:T                   [0 1 0]
3:dim(                [0 0 1]]
4:fill(
5:identity(
6:randM(
7:augment(
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$$Id_{3\times3} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$Id_{m \times n} \cdot \overset{A}{(m \times n)} = A$$

$$A \cdot Id_{(n \times n)} = A.$$

The identity matrix has the following properties:

Example
Cost Analysis - The Mundo Candy Company makes three types of chocolate candy: cheery cherry (cc), mucho mocha (mm) and almond delight (ad).

The candy is produced in San Diego (SD), Mexico City (MC) and Managua (Ma) using two main ingredients, sugar (S) and chocolate (C).

Each kilogram of cheery cherry requires 0.5 kg of sugar and 0.2 kg of chocolate.
Each kilogram of mucho mocha requires 0.4 kg of sugar and 0.3 kg of chocolate.
Each kilogram of almond delight requires 0.3 kg of sugar and 0.3 kg of chocolate.

(a) Put this information in a 2x3 matrix.

$$\begin{array}{c} \\ \text{Sug.} \\ \text{Cho.} \end{array} \begin{array}{c} \begin{array}{ccc} cc & mm & ad \end{array} \\ \begin{pmatrix} 0.5 & 0.4 & 0.3 \\ 0.2 & 0.3 & 0.3 \end{pmatrix} \end{array} = A \quad 2 \times 3$$

(b) The cost of 1 kg of sugar is \$3 in San Diego, \$2 in Mexico City and \$1 in Managua. The cost of 1 kg of chocolate is \$3 in San Diego, \$3 in Mexico City and \$4 in Managua.

Put this information into a matrix in such a way that when it is multiplied by the matrix in part (a) it will tell us the cost of producing each kind of candy in each city.

candies

cities ( )

B

$2\times 3 \qquad 3\times 2$

$2\times 3 \overset{?}{=} 2\times 3 \leftarrow$ NO

$3\times 2 \qquad 2\times 3 \leftarrow$ Yes

(3×3)

$$\begin{array}{c|cc} & S & C \\ \hline \text{cities} \begin{array}{c} SD \\ MC \\ M \end{array} & \begin{pmatrix} 3 & 3 \\ 2 & 3 \\ 1 & 4 \end{pmatrix} \end{array} = B$$

$\rightarrow m \times n \qquad n \times p$

$m \times p$

$B \cdot A$

$$= \begin{pmatrix} 3 & 3 \\ 2 & 3 \\ 1 & 4 \end{pmatrix} \times \begin{array}{c} \begin{array}{ccc} cc & mm & ad \end{array} \\ \begin{pmatrix} 0.5 & 0.4 & 0.3 \\ 0.2 & 0.3 & 0.3 \end{pmatrix} \end{array} \overset{?}{=} \begin{pmatrix} 3\times 0.5+3\times 0.2 & 3\times 0.4+3\times 0.3 & _ \\ _ & _ & _ \\ _ & _ & _ \end{pmatrix}$$

$$\begin{array}{c|ccc} & cc & mm & ad \\ \hline \begin{array}{c} SD \\ MC \\ M \end{array} & \begin{pmatrix} 2\cdot 1 & 2\cdot 1 & 1\cdot 8 \\ 1\cdot 6 & 1\cdot 7 & 1\cdot 5 \\ 1\cdot 3 & 1\cdot 6 & 1\cdot 5 \end{pmatrix} \end{array}$$

Matrix multiplication and linear equations:

Example

Write the following system of linear equations as a matrix equation

$$2x-3y=6$$
$$-x+2y=4$$

Answer

$$\overset{2\times 2}{\begin{bmatrix} 2 & -3 \\ -1 & 2 \end{bmatrix}} \overset{2\times 1}{\begin{bmatrix} x \\ y \end{bmatrix}} = \begin{bmatrix} 6 \\ 4 \end{bmatrix}$$

$\rightsquigarrow 2\times 1$

$$\begin{bmatrix} 2x+(-3)\cdot y \\ -1\cdot x+2y \end{bmatrix} \overset{?}{=} \begin{bmatrix} 6 \\ 4 \end{bmatrix}$$